Xin Sun

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Research Interests

My research focuses on **Retrieval-Augmented Generation (RAG)** and **LLM Agent Training**, with two core directions: **(1) RAG & Deep Search**—building reliable, adaptive, and abstention-aware retrieval-augmented systems that know when to retrieve, what to retrieve, how to retrieve, and when to stop or refuse, toward autonomous deep research agents; **(2) LLM Agent Training**—training LLM-based agents through reinforcement learning and post-training to perform multi-turn exploration, tool use, structured reasoning, fine-grained credit assignment, and learning from environment feedback. I am currently an **Ant Star Intern** with the Machine Intelligence Division at Ant Group. During my Ph.D. study, I have published **7 first-/co-first-author papers in top-tier conferences/journals (6 CCF-A / TH-CPL A and 1 CCF-B)**, with **2 additional co-first-author papers under review** at AAAI 2027.

Education

University of Science and Technology of China (USTC) Hefei, China

Ph.D. Candidate in Control Science and Engineering *Joint Program with CASIA* *2022 – 2027 (Expected)*

- National Laboratory of Pattern Recognition (NLPR), Institute of Automation, Chinese Academy of Sciences (CASIA)
- Advisors: Prof. Liang Wang (IEEE Fellow) and Assoc. Prof. Qiang Liu
- Honors: BYD PhD Scholarship (Oct. 2026); National Scholarship for Graduate Students (2024); USTC First-Class Academic Scholarship (2022–2025, three consecutive years)

Shanghai Jiao Tong University (SJTU) Shanghai, China

B.Eng. in Computer Science and Technology *2018 – 2022*

- Honors: National Encouragement Scholarship (2019–2022); SJTU Academic Scholarship

Publications

▶ First-Author Publications

[1] **KBQA-R1: Reinforcing Large Language Models for Knowledge Base Question Answering**
Xin Sun, Zhongqi Chen, Xing Zheng, Qiang Liu, Shu Wu, Bowen Song, Zilei Wang, Weiqiang Wang, Liang Wang
ICML 2026 (CCF-A) WeChat article (Chinese) Formulates knowledge base question answering as a multi-turn MDP where models adapt their reasoning strategies through closed-loop interaction with the KB; introduces RRS for RL cold start and RRGC for sparse rewards, improving accuracy while achieving 26× higher inference efficiency than MCTS-based methods.

[2] **GAPD: Gold-Action Policy Distillation for Agentic Reinforcement Learning in Knowledge Base Question Answering**
Xin Sun, Zhongqi Chen, Xing Zheng, Qiang Liu, Shu Wu, Bowen Song, Zilei Wang, Weiqiang Wang, Liang Wang
EMNLP 2026, Findings (TH-CPL A) Addresses sparse supervision in outcome-only RL for agentic KBQA by proposing a gold-action-conditioned self-teacher and Entity-Anchor Matching, converting gold actions into safely aligned token-level policy guidance while preserving the original GRPO objective.

[3] **EviSD: Evidence-Conditioned Self-Distillation for Search-Augmented Agents**
Jianan Xie*, **Xin Sun***, Zhongqi Chen, Xing Zheng, Shu Wu, Bowen Song, Liang Wang
Under Review at AAAI 2027 *arXiv:2608.01359* **Equal contribution* Proposes evidence-conditioned self-distillation that uses supporting evidence and golden answers as privileged information for action-level credit assignment, achieving the best macro-average EM across seven QA benchmarks while modulating only 6.7%–15.1% of response tokens.

[4] **HopRefusalBench: Diagnosing Refusal Failures in Search-Augmented Agents for Multi-Hop Reasoning**
Jianan Xie*, **Xin Sun***, Zhongqi Chen, Xing Zheng, Qiang Liu, Bowen Song
Under Review at AAAI 2027 *arXiv:2608.01358* **Equal contribution* Introduces an 889-question benchmark crossing three causes of unanswerability with root, middle, and terminal topologies; evaluation of ten frontier models finds a best target-aware correct halting rate of only 42.9%, exposing persistent hallucination and search-budget exhaustion.

[5] **Divide-Then-Align: Honest Alignment based on the Knowledge Boundary of RAG**
Xin Sun, Jianan Xie, Zhongqi Chen, Qiang Liu, Shu Wu, Yuehe Chen, Bowen Song, Zilei Wang, Weiqiang Wang, Liang Wang
ACL 2025, Main Conference (CCF-A) WeChat article (Chinese) Defines a four-quadrant knowledge boundary for RAG based on parametric and retrieved knowledge, constructs quadrant-specific preference data, and performs honest alignment with DPO, SFT, and auxiliary classification objectives to teach models to refuse when knowledge is insufficient.

[6] **Predict the Retrieval! Test Time Adaptation for Retrieval Augmented Generation**
Xin Sun, Zhongqi Chen, Qiang Liu, Shu Wu, Bowen Song, Weiqiang Wang, Zilei Wang, Liang Wang
ICASSP 2026 (CCF-B) Constructs self-supervised prefix-suffix prediction tasks from retrieved documents and dynamically updates model parameters at inference time, enabling RAG systems to adapt to specialized domains without labeled data.

[7] **DIVE: Subgraph Disagreement for Graph Out-of-Distribution Generalization**
Xin Sun, Liang Wang, Qiang Liu, Shu Wu, Zilei Wang, Liang Wang
KDD 2024 (CCF-A) Learns environment-invariant graph representations via subgraph disagreement, improving graph neural network generalization under distribution shifts.

[8] **Noise-Robust Semi-Supervised Learning for Distant Supervision Relation Extraction**
Xin Sun, Qiang Liu, Shu Wu, Zilei Wang, Liang Wang
EMNLP 2023, Findings (TH-CPL A) Improves distant supervision relation extraction by designing a noise-robust semi-supervised learning framework for noisy labels.

[9] **Uncertainty Calibration for Counterfactual Propensity Estimation in Recommendation**
Wenbo Hu*, **Xin Sun*** **(First Student Author)**, Qiang Liu, Liang Wang
IEEE TKDE (CCF-A, SCI Q1) Improves counterfactual propensity estimation for recommendation through uncertainty calibration, enhancing conversion-rate prediction accuracy and robustness.

▶ Co-authored Publications

[10] **ToolWeaver: Weaving Collaborative Semantics for Scalable Tool Use in Large Language Models**
Bowen Fang, Qiang Liu, Yesheng Liu, Jiabing Yang, **Xin Sun**, Shu Wu, Baole Wei, Liang Wang
ICLR 2026 (CAAI-A) Enables scalable tool selection and composition for LLMs by weaving collaborative semantics over large tool sets.

[11] **Multimodal Adaptive Retrieval Augmented Generation through Internal Representation Learning**
Ruoshuang Du, **Xin Sun**, Qiang Liu, Bowen Song, Zhongqi Chen, Weiqiang Wang, Liang Wang
IJCNN 2026 (CCF-C) Enables adaptive retrieval-augmented generation in multimodal settings through internal representation learning.

[12] **Pin-Tuning: Parameter-Efficient In-Context Tuning for Few-shot Molecular Property Prediction**
Liang Wang, Qiang Liu, Shaozhen Liu, **Xin Sun**, Shu Wu, Liang Wang
NeurIPS 2024 (CCF-A) A parameter-efficient in-context tuning method for few-shot molecular property prediction.

[13] **GSLB: The Graph Structure Learning Benchmark**
Zhixun Li, Liang Wang, **Xin Sun**, Yifan Luo, Yanqiao Zhu, Dingshuo Chen, Yingtao Luo, Xiangxin Zhou, Qiang Liu, Shu Wu, Liang Wang, Jeffrey Xu Yu
NeurIPS 2023 (CCF-A) The first comprehensive graph structure learning benchmark, integrating 16 GSL algorithms and 20 graph datasets.

Experience

Ant Group Hangzhou, China
Machine Intelligence Division – Ant Star Intern (since May 2026) *Nov. 2024 – Present*
- Built a Trustworthy RAG research line and led multiple retrieval-augmented generation projects, including honest alignment for RAG (ACL 2025), multimodal adaptive RAG (IJCNN 2026), test-time adaptation for RAG (ICASSP 2026), reinforcement learning for KBQA with KBQA-R1 (ICML 2026), fine-grained credit assignment for search agents, and multi-hop refusal reliability evaluation (AAAI 2027 under review).

ByteDance Beijing, China
AML – Machine Learning Engineer Intern *Feb. 2022 – Aug. 2022*
- Contributed to the development of an AutoML platform for multivariate time-series forecasting.
- Conducted research on causal inference for recommendation systems and proposed an uncertainty calibration method for counterfactual propensity estimation, later published in IEEE TKDE (CCF-A).

Personal Interests

Badminton Skiing Football (FC Barcelona) Board Games

Academic Service

Reviewer: ICML 2026, NeurIPS 2025, ACL 2025, EMNLP 2025, KDD 2024